

सुरक्षा अलर्ट / SAFETY ALERT

OISD/SA/2025-26/PL/01

Dated: 15/04/2025

Introduction:

Title: Incident during unloading of rebar bundle from trailer.
Location: Polypropylene plant construction site.
Loss/ Outcome: Fatality of the contract worker.

Brief of Incident:

The job involved unloading 10 mm rebar bundles from a trailer at a construction site using a 15-ton capacity Farana crane (boom-end hook capacity: 7.6 tons). Each bundle was 12 meters long, weighed approximately 4 tons, and was tied with metal strips at three points. Three fitters were engaged in the unloading operation, excluding the crane operator—two standing on the trailer and one on the ground acting as a signalman.

Out of ten bundles, seven were successfully unloaded when the accident occurred while handling the eighth bundle. The metal strip securing the rebar snapped, causing it to loosen and become entangled with other bundles. As the crane attempted to lift the bundle, it was partially placed on the trailer's side wall while corrective action was being taken. A fitter on the trailer attempted to secure the loose rebars by placing a sling around them. However, the bundle suddenly shifted, pushing the fitter off the trailer and rebars fell on him. He was immediately rescued by co-workers, who lifted the bundle using the crane. He was rushed to a nearby civil hospital, however, the worker succumbed to his injuries.



Actual site pictures

Observations:

- Although rebar unloading was considered a routine task, no SOP or job-specific procedure was available.
- There was no designated signaller or helper assigned to the crane operator, leading to unsynchronized communication between the fitter (acting as a signaller) and the crane operator.
- The fitters engaged in the unloading activity had limited experience and were approximately 25-26 years old.
- The crane operator was a habitual user of earphones and mobile phones during lifting, loading, and unloading activities.
- The web slings used for lifting the rebars were in poor condition; however, the accident was not caused by sling failure.
- A Hazard Identification and Risk Assessment (HIRA) was available for general loading and unloading activities, but specific hazards related to rebar handling were not identified.
- A work permit was issued to the contractor for multiple activities at the construction site, but it was a generic permit.
- Job Safety Analysis (JSA) was not conducted to identify potential hazards associated with the activity.
- Although generalized toolbox talks were regularly conducted, no job-specific toolbox talk was provided.

Root Cause:

Although rebar unloading was a regular activity, the accident happened due to

- i. Non-availability of Job-specific procedure for the task
- ii. Failure to carry out risk assessment for snapping of metal strips securing re-bars
- iii. Untrained workers and casual approach to work
- iv. Lack of supervision

Recommendations:

- Site-specific job procedures, work practices shall be developed for the safe execution of jobs like handling and lifting rebars as per clause No. 6.1 of OISD-STD-105.
- Job-specific toolbox talks shall be conducted as per clause No. 6.3.3 of OISD-STD-105.
- Training and competencies of contractor workers shall be ensured so that work is executed responsibly and safely as per clause No. 7.4 of OISD-GDN-206.
- Awareness about unsafe acts (UA) and unsafe conditions (UC) among contract workers shall be improved through training as per OISD-STD-154. Reporting of UA/UC by contractor workforce should be encouraged in line with clause No. 7.5.3 of OISD-GDN-206.
- Use of earphones and mobile phones shall be restricted at the construction site.
- All lifting slings, chains, etc., shall be physically checked every time before use.
- Bundling of rebars shall be ensured with sufficiently strong metal strips so that they do not snap during transportation.
- When handling flexible rebars, no worker shall stand within the boom movement area.
- Guide ropes shall be used during lifting activities to prevent unsafe movement.
- Adequacy of Hydra/crane shall be checked before lifting/unloading of rebars.

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